

Lesson 2.3: Regret

Decision-making under uncertainty

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Rationality criteria

- is *Maximin* a ‘rational’ decision rule?

A	4
B	3

	s_1	s_2
A	4	2
B	3	2

	s_1	s_2	min
A	4	2	2
B	3	2	2

	s_1	s_2	min
A	4	5	4
B	3	5	3

Best response

Recall the fundraiser example:

	d	w
S	120	85
F	150	75

- ▶ *best response* on dry days is F
- ▶ best response on wet days is S

Regret

	d	w
S	120	85
F	150	75

Definition (Regret)

If the actual state were s , then the *regret* associated with choosing action A is measured by the difference between the payoff under the best response in s and the payoff under A.

	d	w
S	120	85
F	150	75
Max	150	85

	d	w
S	30	0
F	0	10

Minimax regret

- choose action with *minimal maximum regret*

	d	w	Max
S	30	0	30
F	0	10	10

- *Minimax regret* chooses F

Example

	s_1	s_2	s_3
A_1	6	0	4
A_2	2	5	1
A_3	4	3	2

	s_1	s_2	s_3
A_1	6	0	4
A_2	2	5	1
A_3	4	3	2
Max	6	5	4

	s_1	s_2	s_3	Max
A_1	0	5	0	5
A_2	4	0	3	4
A_3	2	2	2	2

- *Minimax regret* chooses A_3